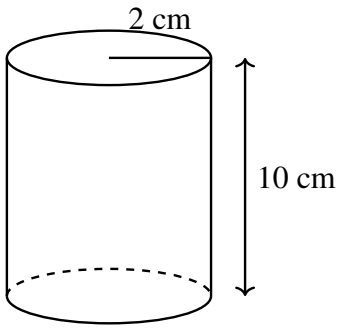
Think Academy

Bonus Lesson 2

3-D Geometry Practice

1. A closed cylinder has a height of 10 cm and a radius of 2 cm.



(i) Calculate the volume of the cylinder. Give your answer correct to 2 decimal places.

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(ii) Calculate the total surface area of the cylinder. Give your answer correct to 2 decimal places.

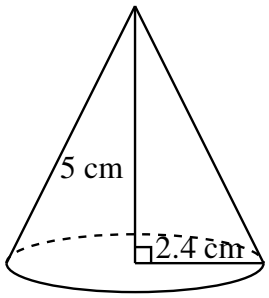
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2. A solid cone has a radius of 2.4 cm and a perpendicular height of 5 cm.



Calculate the following, giving your answers correct to 2 decimal places.

(i) The slant height of the cone.

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(ii) The total surface area of the cone.

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(iii) What fraction of the volume of a cylinder with the same radius and perpendicular height is the volume of this cone?

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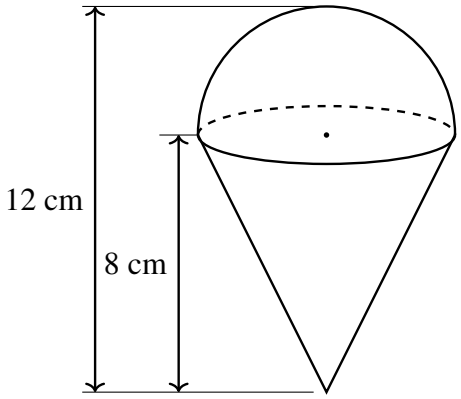
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3. The diagram below shows a solid consisting of a hemisphere stacked on top of a cone.

   The total height of the solid is 12 cm, and the perpendicular height of the cone is 8 cm.



   Calculate the volume of the solid, giving your answer correct to 2 decimal places.

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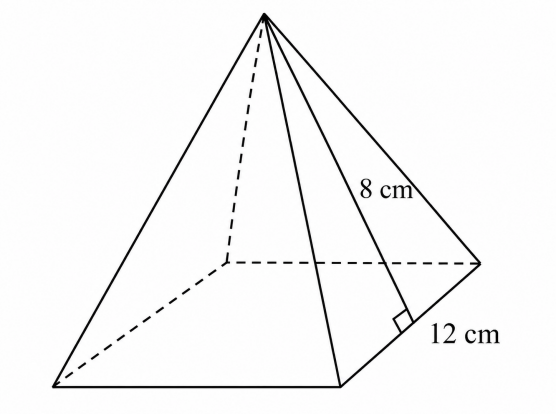
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4. A square-based pyramid has a base edge of 12 cm and a slant height of 8 cm, as shown below.



(a) Calculate the total surface area of the pyramid.

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(b) Calculate the volume of the pyramid, giving your answer correct to 1 decimal place.

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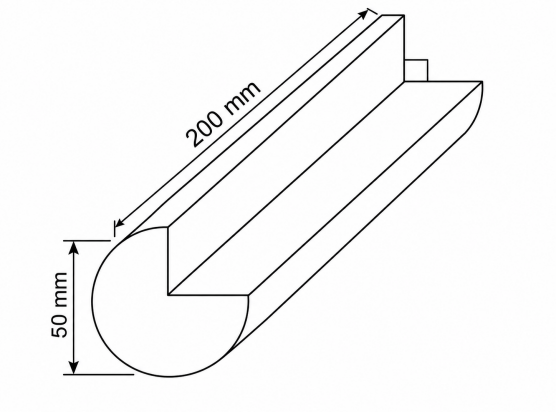
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5. The diagram below shows a solid prism formed by removing one-quarter of a cylinder.

   The cylinder has a diameter of 50 mm and a length of 200 mm.



   Calculate the following:

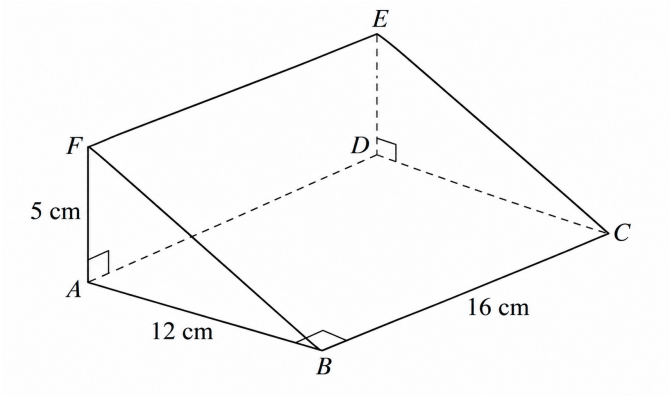
   (i) The volume of the prism.

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   (ii) The total surface area of the prism.

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6. A right triangular prism is shown in the diagram below, with $AB = 12$ cm, $BC = 16$ cm and $AF = 5$ cm.



(i) Calculate the volume of the prism.

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(ii) Calculate the total surface area of the prism.

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(iii) Determine the size of the angle between the diagonals AC and CF. Give your answer correct to the nearest degree.

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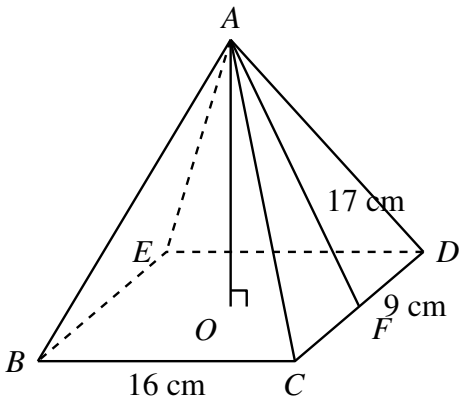
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7. In the rectangular pyramid shown below, $BC = 16$ cm, $CD = 9$ cm and $AF = 17$ cm.



(i) Write down the length of OF.

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(ii) Calculate AO, the perpendicular height of the pyramid.

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(iii) Hence, calculate the volume of the pyramid.

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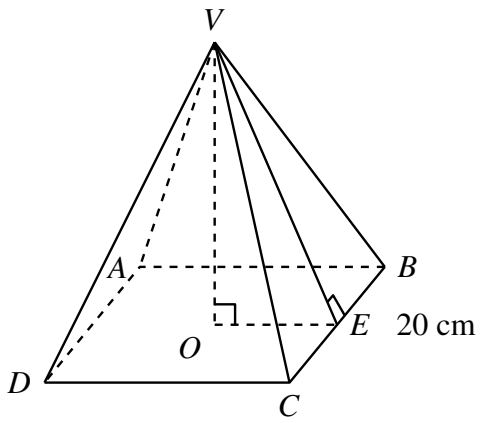
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8. The base edge BC of a right square-based pyramid $VABCD$ is 20 cm. The apex V is directly above the centre O of the square base, and E is the midpoint of BC.



(a) Given that the total surface area of the pyramid is 1600 cm^2, calculate the perpendicular height VE of a triangular face.

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(b) Find the perpendicular height VO of the pyramid. Leave your answer in exact form.

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(c) Hence, calculate the volume of the pyramid.

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(d) A second similar pyramid has a base edge of $15a$ cm. Find the volume of the second pyramid in terms of a.

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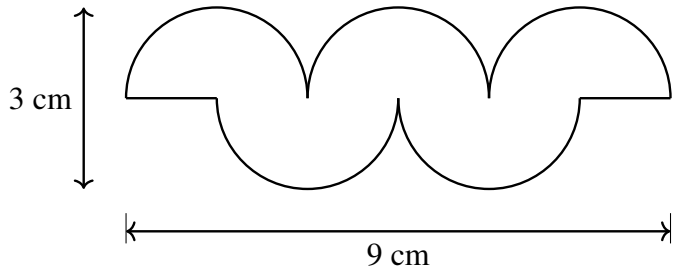
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9. Badges have been designed as shown below.



All the curves in the design are semicircles with the same radius. The badge has a uniform thickness of 0.3 cm.

Calculate the volume of the badge, correct to 1 decimal place.

A. 5.3 cm^3

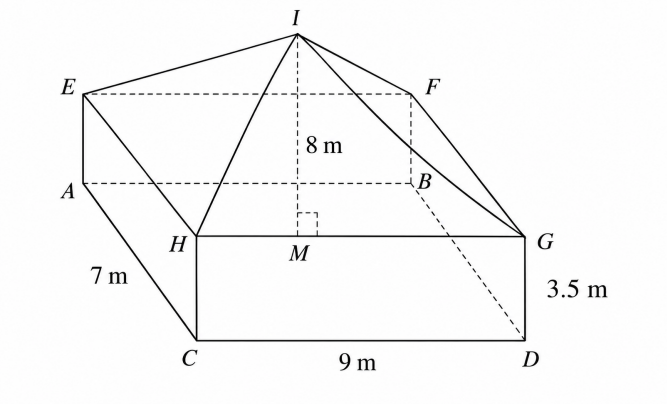
B. 10.6 cm^3

C. 21.2 cm^3

D. 42.4 cm^3

10. The diagram shows a cuboid and a pyramid. The apex of the pyramid, I, is directly above the centre, M, of $ABDC$.

The cuboid has a length of 9 m, a width of 7 m and a height of 3.5 m. The perpendicular height IM of the pyramid is 8 m.



(a) Calculate the angle between the line DI and the plane $ABDC$.

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(b) Calculate the angle between the planes EHI and $ACHE$.

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